



CSCE 633-700: Machine Learning

Linear regression

Overview

Linear Regression

- Example
- Representation, Evaluation
- Optimization: Closed form solution via Ordinary Least Squares
- Optimization: Numerical solution via Gradient Descent
 - General gradient descent
 - Gradient descent for linear regression (batch, stochastic, mini-batch)
- Non-linear basis function for regression & Overfitting

[Parts of these slides have been adapted from K. Murphy (Machine Learning: A probabilistic perspective), Dr. Andrew Ng's Machine Learning course at Coursera, and CSCI567 Machine Learning (USC, Drs. Sha & Liu)]

Linear Regression: Optimization

- Our goal is to find the solution $\mathbf{w}^*$ to minimize the prediction error:
 $\mathbf{w}^* = \arg \min_{\mathbf{w}} RSS(\mathbf{w})$
- The cost function has a 1st order derivative, which if we set to zero, we can find a **closed-form solution**

We will first expand the vector/matrix expression of $RSS(\mathbf{w})$:

$$\begin{aligned} RSS(\mathbf{w}) &= (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w}) = \mathbf{y}^T \mathbf{y} - \mathbf{y}^T (\mathbf{X}\mathbf{w}) - (\mathbf{X}\mathbf{w})^T \mathbf{y} + (\mathbf{X}\mathbf{w})^T (\mathbf{X}\mathbf{w}) \\ &= \mathbf{y}^T \mathbf{y} - 2(\mathbf{X}\mathbf{w})^T \mathbf{y} + (\mathbf{X}\mathbf{w})^T (\mathbf{X}\mathbf{w}) = \mathbf{y}^T \mathbf{y} - 2\mathbf{w}^T (\mathbf{X}^T \mathbf{y}) + \mathbf{w}^T (\mathbf{X}^T \mathbf{X}) \mathbf{w} \end{aligned}$$

We then compute the first-order derivative $\frac{\partial RSS(\mathbf{w})}{\partial \mathbf{w}}$:

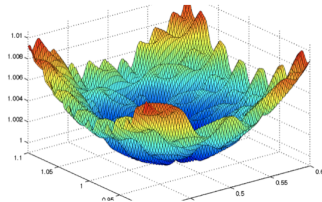
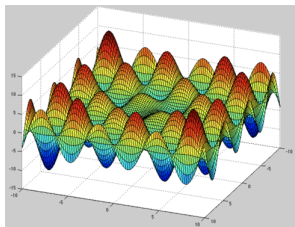
$$\frac{\partial RSS(\mathbf{w})}{\partial \mathbf{w}} = -2(\mathbf{X}^T \mathbf{y}) + 2(\mathbf{X}^T \mathbf{X}) \mathbf{w}$$

We set the first-order derivative to 0 and solve with respect to $\mathbf{w}$:

$$\frac{\partial RSS(\mathbf{w})}{\partial \mathbf{w}} = 0 \Rightarrow -2(\mathbf{X}^T \mathbf{y}) + 2(\mathbf{X}^T \mathbf{X}) \mathbf{w} = 0 \Rightarrow (\mathbf{X}^T \mathbf{X}) \mathbf{w} = (\mathbf{X}^T \mathbf{y}) \Rightarrow \mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Linear Regression: Optimization

Why should convexity be a problem in optimization?



Loss functions might have more than one local optima (minima or maxima)

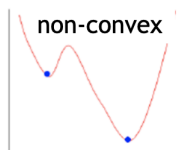
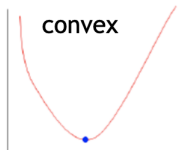
Linear Regression: Optimization

Theorem

Consider an optimization problem:

$\min f(\mathbf{x})$ s.t. $\mathbf{x} \in \Omega$ where f is a convex function and Ω is a convex set.

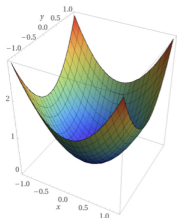
Then any local minimum is also a global minimum



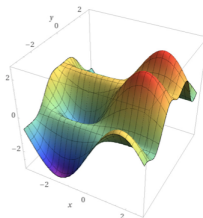
Linear Regression: Optimization

The second order derivative test

If the Hessian matrix $\mathbf{H}_f$ of function $f(\mathbf{x})$ is positive semi-definite, then f is convex, i.e., $\mathbf{u}^T \mathbf{H}_f \mathbf{u} \geq 0$ for every $\mathbf{u}$



convex



non-convex

Gradient vector and Hessian matrix

Gradient vector:

$$f : \mathbb{R}^D \rightarrow \mathbb{R}, \nabla f(\mathbf{x}) = \left[\frac{\partial f(\mathbf{x})}{\partial x_1}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_D} \right] \in \mathbb{R}^D$$

$$\text{e.g., } f(\mathbf{x}) = f(x_1, x_2) = x_1^2 + x_2^2, \nabla f(\mathbf{x}) = [2x_1, 2x_2] \in \mathbb{R}^D$$

Hessian matrix:

$$f : \mathbb{R}^D \rightarrow \mathbb{R}, \mathbf{H} = \nabla \left((\nabla f(\mathbf{x}))^T \right) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_D} \\ \vdots & & \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_D \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_D^2} \end{bmatrix} \in \mathbb{R}^{D \times D}$$

$$\text{e.g., } f(\mathbf{x}) = f(x_1, x_2) = x_1^2 + x_2^2 \in \mathbb{R},$$

$$\mathbf{H} = \nabla \left([2x_1, 2x_2]^T \right) = \nabla \left(\begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} \right) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

Linear Regression: Optimization

Convexity of optimization criterion

$$RSS(\mathbf{w}) = \mathbf{y}^T \mathbf{y} - 2\mathbf{w}^T (\mathbf{X}^T \mathbf{y}) + \mathbf{w}^T (\mathbf{X}^T \mathbf{X}) \mathbf{w}$$

$$\frac{\partial RSS(\mathbf{w})}{\partial \mathbf{w}} = -2(\mathbf{X}^T \mathbf{y}) + 2(\mathbf{X}^T \mathbf{X}) \mathbf{w}$$

$$\mathbf{H}_{RSS(\mathbf{w})} = \frac{\partial^2 RSS(\mathbf{w})}{\partial \mathbf{w}^2} = \frac{\partial}{\partial \mathbf{w}} \left(\frac{\partial RSS(\mathbf{w})}{\partial \mathbf{w}} \right) = \frac{\partial}{\partial \mathbf{w}} \left(-2(\mathbf{X}^T \mathbf{y}) + 2(\mathbf{X}^T \mathbf{X}) \mathbf{w} \right) = 2(\mathbf{X}^T \mathbf{X})$$

For every $\mathbf{u} \in \mathbb{R}^D$ we have (by applying the transpose product rule and the definition of l_2 -norm):

$$\mathbf{u}^T \mathbf{H}_{RSS(\mathbf{w})} \mathbf{u} = 2\mathbf{u}^T (\mathbf{X}^T \mathbf{X}) \mathbf{u} = 2\mathbf{u}^T \mathbf{X}^T \mathbf{X} \mathbf{u} = 2(\mathbf{X} \mathbf{u})^T \mathbf{X} \mathbf{u} = 2\|\mathbf{X} \mathbf{u}\|_2^2 \geq 0$$

Therefore the Hessian $\mathbf{H}_{RSS(\mathbf{w})}$ of the RSS error is positive semi-definite, thus $RSS(\mathbf{w})$ is convex and any local optima is a global minimum.

Therefore the solution $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \in \mathbb{R}^D$ is a global minimum of the RSS error of the linear regression problem.